

Unleash Your Inner Monster: A Demonstration of High-Fidelity Human to Non-Human Voice Conversion

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Abstract

Non-human monsters in massive multiplayer online role-playing games (MMORPGs) are essential for immersive game environments and significantly enhance player engagement. However, producing high-quality monster sounds demands considerable time and financial resources. AI-driven voice conversion offers a potential solution, but existing models rely on standard human voice training and human-centric audio features, limiting their ability to generate realistic non-human voices. This study presents a human-to-non-human (H2NH) voice conversion model designed to address these challenges. The model effectively generated high-quality non-human sounds by recording the voices of participants and converting them in real-time into selected monster vocalizations.

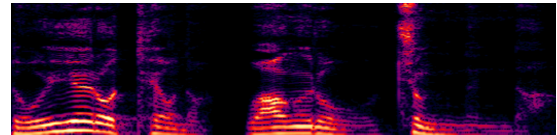
Index Terms: non-human voice conversion, monster sound

1. Introduction

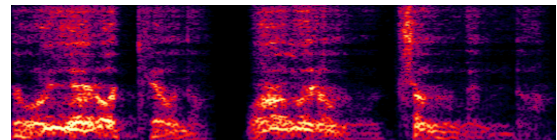
The global gaming market expanded rapidly during the COVID-19 pandemic as demand for stay-at-home, non-face-to-face entertainment increased [1]. Recent massive multiplayer online role-playing games (MMORPGs), such as Lineage W, now feature high-quality 3D graphics and 48 kHz high-fidelity sound to improve realism. Medieval fantasy-themed games, inspired by Dungeons & Dragons [2], often incorporate fictional monsters such as dragons and orcs to enrich the virtual world. Achieving such immersive environments requires extensive creativity not only from graphic designers shaping a monster's appearance but also from sound designers crafting its voice—an effort that incurs significant costs. This study proposes a voice conversion model that transforms human voices into non-human timbres, reducing production costs. Section 2 examines the characteristics of monster sounds. Section 3 reviews recent voice conversion technologies and their limitations. Section 4 introduces the proposed model, while Section 5 details the implementation and demonstration of the model. Section 6 presents the conclusions.

2. The Monster Sound

Monster sounds in games occupy a unique position between voice and sound effects. While they convey linguistic information, making them voice-like, they also require extensive filtering to achieve textures that define the identity of the monster, classifying them as a specialized type of sound effect. Additionally, they incorporate elements rarely addressed by conventional speech technology, such as unvoiced sound (e.g., breathing) and extreme vocal expressions like screaming. Therefore, producing high-quality monster voices demands exceptional performances from voice actors and extensive postproduction work,



(a) Source audio of "Born a slave, died a king." performed in Korean.



(b) Result audio of voice conversion into an Orc voice.

Figure 1: Spectrogram comparison of a human voice and an Orc-converted monster sound.

resulting in higher time and financial costs than other in-game audios.

The production process generally involves three stages: (1) defining the timbre of the target monster, (2) hiring a professional voice actor capable of approximating that timbre, and (3) applying effects and modulation to refine the recorded voice. The timbre of a monster is largely determined by its origin, size, and sex. Employing skilled voice actors is costly because the vocal range required is broader than standard narration and imposes greater strain. Finally, the recorded "seed voice" undergoes post-processing to achieve the final monster timbre, typically using specialized tools, such as Dehumaniser¹, a Digital Audio Workstation (DAW) plug-in. Producing high-quality monster sounds requires trained personnel to operate these tools effectively. Advanced voice conversion is anticipated to reduce these time and financial costs.

3. Related Works

Voice conversion for ordinary speech has advanced to the point where defenses against adversarial attacks must be considered [3]. Additionally, techniques such as singing voice conversion using latent diffusion models [4] achieve highly accurate timbral reproduction, with users frequently sharing redubbed singing videos on social media for entertainment. These models rely on speech features that capture human vocal characteristics and are trained exclusively on human-speech datasets to ensure generalization. However, monster voices, such as those inspired by birds or tigers, span a wider frequency range than human voices and contain intricate details shaped by various

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¹<https://www.krotosaudio.com/dehumaniser2/>

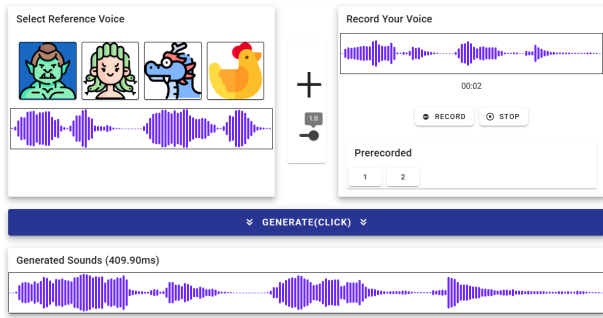


Figure 2: Web UI for the demonstration of the H2NH voice conversion model.

audio effects. Consequently, voice conversion models trained on standard human speech struggle to reproduce the timbre of reference monster voices accurately. No publicly available technology exists to convert human voices into monster voices.

4. Proposed Method

The model employed a 44.1 kHz sampling rate, STFT with a 20 ms window and 5 ms hop length, and Mel spectrograms covering 0–22.05 kHz to accommodate a broad frequency range and preserve fine temporal details. A reference encoder extracted the global style from Mel spectrograms, while self-supervised perturbations captured linguistic features across pitch and formant variations. Given the challenges in reliably estimating f_0 for non-human sounds, frame-level energy is the main prosodic cue.

The proposed architecture consisted of a CVAE-based posterior encoder, decoder, prior encoder, and flow module, with two key enhancements: incorporating a 5 ms hop-length linear spectrogram input and applying the style vector exclusively within the prior and flow components. The prior encoder integrated style embeddings, energy inputs, and SSL-derived linguistic features, estimated latent distribution parameters, and refined the sampled latent variable using a flow module.

Training employed a combined generator and discriminator losses, including KL divergence, feature matching, LS-GAN adversarial loss, and reconstruction loss. The discriminator was trained separately using adversarial loss. The model incorporated FDRl with multiple hop lengths and a DAC discriminator to enhance temporal resolution to better handle the transient-rich nature of non-human audio. Figure 1 shows the monster sound converted into an Orc voice from the original human voice using the proposed model.

The content summarized above is to be published at Interspeech 2025 [5]. Comparative results against state-of-the-art voice conversion models such as DDDM-VC [6] are included. Conversion samples, generated using a public speech corpus as the source voice audio, are available at the following URL: <https://nc-ai.github.io/speech/publications/nonhuman-vc>.

5. Demonstration

The proposed model is packaged to run independently within a Linux virtual machine environment, enabling users to interact with it via a Windows-based web interface on the local host. Figure 2 illustrates the Web UI used for the demonstration. Users begin by selecting a reference monster voice from available target voices. Subsequently, they record their voices as

the source input using a Web Audio API-based AudioWorklet, facilitating real-time browser access and playback for review. Once the recording is completed, clicking the “convert” button transmits both the selected reference monster voice and the recorded source voice to the H2NH voice conversion model running on localhost. The converted voice is returned to the web interface upon the transformation and played immediately. This web-based implementation provides an intuitive and user-friendly experience, requiring no complex setup while delivering instant feedback on transformation results. Consequently, users can enjoy an engaging and interactive experience in this demonstration.

6. Conclusion

This study introduces an efficient voice conversion model that streamlines the production of high-quality monster voices in MMORPGs, addressing the substantial costs and specialized processes associated with conventional methods. The human-to-non-human (H2NH) voice conversion model uses an expanded frequency range and enhanced temporal resolution to accurately capture and transform the intricate characteristics of non-human voices. A real-time demonstration system was implemented to validate the ability of the model to convert the voice of a speaker into a realistic monster-like timbre. The proposed approach can significantly reduce the time and expenses required to produce monster voices in game development. Future research will focus on extending the capabilities of the model to a broader range of monster and speech domains, to develop a general-purpose voice conversion system that maintains consistently high performance.

7. Acknowledgements

This research was supported by Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2024 (Project Name: Development of Co-Pilot technology for automatic completion of generative AI-based 3D Webtoon, Project Number: RS-2024-00400004, Contribution Rate: 100%)

8. References

- [1] D. Şener, T. Yalçın, and O. Gulseven, “The impact of covid-19 on the video game industry,” *Available at SSRN 3766147*, 2021.
- [2] G. Gygax and D. Arneson, *Dungeons & dragons. Tactical Studies Rules* Lake Geneva, WI, 1974, vol. 19.
- [3] J. Gao, H. Li, Z. Zhang, and Z. Wu, “Black-box adversarial defense against voice conversion using latent space perturbation,” in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [4] S. Chen, Y. Gu, J. Zhang, N. Li, R. Chen, L. Chen, and L. Dai, “Ldm-svc: Latent diffusion model based zero-shot any-to-any singing voice conversion with singer guidance,” in *Proc. Interspeech 2024*, 2024, pp. 2770–2774.
- [5] M.Kang, S.Lee, C.Lee, and N.Cho, “When humans growl and birds speak: High-fidelity voice conversion from human to animal and designed sounds,” in *Proc. Interspeech*, 2025 (in press) DOI: 10.21437/Interspeech.2025-433.
- [6] H. Choi, S. Lee, and S. Lee, “Dddm-vc: Decoupled denoising diffusion models with disentangled representation and prior mixup for verified robust voice conversion,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, 2024, pp. 17 862–17 870.